



CSS Media Queries

Master responsive design with media queries for all devices

What are CSS Media Queries?

CSS Media Queries allow you to apply different styles to your website based on device characteristics like screen width, height, orientation, and other features. They're the foundation of responsive web design, enabling your site to look great on everything from small phones to large desktop monitors.

Media queries let you create layouts that adapt automatically to the user's device, without requiring multiple separate websites or applications.

Browser Support: All modern browsers support media queries. IE 9+ supports them. Mobile browsers support media queries natively.

Media Query Syntax

Basic Syntax

```
Syntax: @media (condition) { /* CSS rules */ }

Example:
@media (max-width: 768px) {
  .sidebar { display: none; }
  .main { width: 100%; }
}
```

Media Query Breakdown

```
@media (min-width: 600px) and (max-width: 999px) {
  /* Styles for tablets */
}

@media screen and (orientation: landscape) {
  /* Styles for landscape orientation */
}

@media only screen and (color) {
  /* Styles for color screens only */
}
```

Media Features & Conditions

Width-Based Queries (Most Common)

```
/* Exactly 480px */
@media (width: 480px) { }

/* 480px or narrower */
@media (max-width: 480px) { }

/* 480px or wider */
@media (min-width: 480px) { }

/* Between 480px and 768px */
@media (min-width: 480px) and (max-width: 768px) { }
```

Height-Based Queries

```
/* Viewport height 600px or taller */
@media (min-height: 600px) { }

/* Viewport height 400px or shorter */
@media (max-height: 400px) { }

/* Exact height */
@media (height: 768px) { }
```

Orientation

```
/* Portrait orientation (height > width) */
@media (orientation: portrait) { }

/* Landscape orientation (width > height) */
@media (orientation: landscape) { }
```

Device Type

```
/* Screen devices only */
@media screen { }

/* Print media */
@media print { }

/* Speech/screen reader */
@media speech { }
```

Color & Display Features

```
/* Color screen (supports color) */
@media (color) { }

/* Monochrome screen (black & white) */
@media (monochrome) { }

/* High DPI screen (retina) */
@media (min-resolution: 2dppx) { }
```

Aspect Ratio

```
/* 16:9 aspect ratio */
@media (aspect-ratio: 16/9) { }

/* Between 4:3 and 16:9 */
@media (min-aspect-ratio: 4/3) and (max-aspect-ratio: 16/9) { }
```



Interactive Media Query Demos

Resize your browser window to see how the layout adapts to different screen sizes

Responsive Grid Layout

This grid automatically changes from 1 column (mobile) – 2 columns (tablet) – 3 columns (desktop)

Mobile
1 Column

Tablet
2 Columns

Desktop
3 Columns

Common Breakpoints

Standard Breakpoints (Mobile First)

Mobile: 320px - 480px

Tablet: 481px - 768px

Desktop: 769px+

Bootstrap Breakpoints

```
/* Extra small (mobile) */
@media (max-width: 575px) { }

/* Small (landscape mobile/tablet) */
@media (min-width: 576px) { }

/* Medium (tablet) */
@media (min-width: 768px) { }

/* Large (small desktop) */
@media (min-width: 992px) { }

/* Extra large (desktop) */
@media (min-width: 1200px) { }
```

Tailwind CSS Breakpoints

```
sm: 640px
md: 768px
lg: 1024px
xl: 1280px
2xl: 1536px
```

Practical Examples & Patterns

1. Mobile-First Approach

```
/* Mobile styles (default) */
.container { width: 100%; }
.sidebar { display: none; }

/* Tablet and up */
@media (min-width: 768px) {
  .container { width: 95%; margin: 0 auto; }
}

/* Desktop and up */
@media (min-width: 1024px) {
  .container { display: grid; grid-template-columns: 250px 1fr; }
  .sidebar { display: block; }
}
```

2. Hide/Show Elements by Device

```
/* Hide desktop nav on mobile */
.desktop-nav { display: none; }
.mobile-menu { display: block; }

@media (min-width: 768px) {
  .desktop-nav { display: block; }
  .mobile-menu { display: none; }
}
```

3. Responsive Typography

```
/* Mobile */
h1 { font-size: 24px; }
body { font-size: 14px; }

/* Tablet */
@media (min-width: 768px) {
  h1 { font-size: 32px; }
  body { font-size: 16px; }
}

/* Desktop */
@media (min-width: 1024px) {
  h1 { font-size: 48px; }
  body { font-size: 18px; }
}
```

4. Layout Changes

```
/* Mobile: Single column */
.layout { display: block; }

/* Tablet: 2 columns */
@media (min-width: 768px) {
  .layout { display: grid; grid-template-columns: 1fr 1fr; }
}

/* Desktop: 3 columns */
@media (min-width: 1024px) {
  .layout { grid-template-columns: 1fr 2fr 1fr; }
}
```

5. Print Styles

```
/* Hide elements not needed for print */
@media print {
  .no-print { display: none; }
  body { font-size: 12pt; line-height: 1.5; }
  a { text-decoration: underline; }
  .page-break { page-break-before: always; }
}
```

6. Orientation-Based Styles

```
/* Landscape phone */
@media (max-height: 500px) and (orientation: landscape) {
  header { height: 50px; }
  h1 { font-size: 18px; }
}
```

7. High DPI (Retina) Displays

```
/* Regular display */
.logo { background-image: url('logo.png'); }

/* Retina display (2x) */
@media (-webkit-min-device-pixel-ratio: 2), (min-resolution: 192dpi) {
  .logo { background-image: url('logo@2x.png'); }
}
```

8. Dark Mode

```
/* Light theme (default) */
body { background: white; color: black; }

/* Respect dark mode preference */
@media (prefers-color-scheme: dark) {
  body { background: #1a1a1a; color: #fff; }
}
```

Best Practices

- **Use mobile-first approach:** Start with mobile styles, then add more complex layouts for larger screens.
- **Use consistent breakpoints:** Choose a system (Bootstrap, Tailwind, custom) and stick with it.
- **Test on real devices:** Browser DevTools are helpful, but test on actual phones and tablets.
- **Use relative units:** Use em, rem, %, vw instead of fixed px for better responsive behavior.
- **Don't hide too much content:** Ensure important content isn't completely hidden on mobile.
- **Touch-friendly on mobile:** Make buttons/links large enough (min 44x44px) for touch on mobile.
- **Use logical CSS:** Use CSS Grid and Flexbox instead of floats for modern responsive layouts.
- **Consider performance:** Don't load desktop images on mobile - use responsive images (srcset).
- **Test in landscape too:** Mobile devices in landscape mode have different constraints than portrait.

Advanced Media Query Features

Combining Multiple Conditions

```
/* AND operator - all conditions must be true */
@media (min-width: 768px) and (orientation: portrait) { }

/* OR operator - any condition can be true */
@media screen, print { }

/* NOT operator */
@media not print { }

/* Complex combination */
@media (min-width: 768px) and (max-width: 1024px), (min-width: 1400px) { }
```

User Preference Media Queries

```
/* Reduced motion preference */
@media (prefers-reduced-motion: reduce) {
  * { animation-duration: 0.01ms !important; }
}

/* Dark mode preference */
@media (prefers-color-scheme: dark) { }
```

Future Media Queries

```
/* Scripting support */
@media (scripting: enabled) { }

/* Hover capability */
@media (hover: hover) { }
```

```
/* Pointer capability */
@media (pointer: fine) { }
```

Browser Compatibility

Modern Browsers: Chrome, Firefox, Safari, Edge all support media queries with most common features.

IE 9+: Basic media queries supported. Some newer features like prefers-color-scheme not supported.

Mobile Browsers: All mobile browsers support media queries natively.

IE 8 and below: No media query support. Requires JavaScript polyfill or mobile detection fallback.

Quick Reference

Common Media Query Syntax

```
/* Screen and minimum width */
@media screen and (min-width: 768px) { }

/* Maximum width */
@media (max-width: 767px) { }

/* Between widths */
@media (min-width: 768px) and (max-width: 1023px) { }

/* Orientation */
@media (orientation: landscape) { }

/* Print */
@media print { }

/* Retina */
@media (-webkit-min-device-pixel-ratio: 2), (min-resolution: 192dpi) { }
```

Mobile-First Template

```
/* Mobile (default) */
body { font-size: 14px; }
.container { padding: 10px; }

/* Tablet up */
@media (min-width: 768px) {
  body { font-size: 16px; }
  .container { padding: 20px; }
}

/* Desktop up */
@media (min-width: 1024px) {
  body { font-size: 18px; }
  .container { padding: 40px; }
}
```